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EDUCATION

National University of Singapore *B.Sc. Physics (Honours, Distinction)*

Aug 2024 – Present (expected May 2028)

GPA: 4.36 / 5.00

University of Texas at Austin *Exchange Semester, Department of Physics*

Aug 2026 – Dec 2026 (incoming)

SUMMARY

Year-2 physics undergraduate working toward research in complex and nonlinear systems. Hands-on experience across 15+ laboratory experiments spanning condensed matter, spectroscopy, plasma physics, biophysics, and ion-matter interactions, with quantified results. Comfortable building physical models and simulations from scratch in Python, Julia, and MATLAB.

EXPERIMENTAL PHYSICS

Experimental Physics I (PC2193):

- Measured **carrier type**, **density**, and **mobility** in n- and p-type germanium via the **Hall effect**; cross-validated mobility by independent **magnetoresistance analysis** ($R^2 > 0.99$); characterized the **extrinsic-to-intrinsic transition** at 106°C
- Determined **lattice constants** via **X-ray diffraction** using **Bragg's law**; achieved **sub-2% accuracy** for LiF and **0.24% deviation** for KBr; extracted **Planck's constant** from bremsstrahlung cutoff analysis
- Calibrated the **Helmholtz field constant** through five independent methods (**3.3% uncertainty**); validated **electromagnetic torque scaling laws** ($R^2 > 0.997$)
- Characterized **electron spin resonance** in paramagnetic systems; extracted **g-factors** from frequency-field data ($R^2 = 0.9995$); analyzed **Zeeman splitting**
- Measured **Gaussian beam propagation**; determined **beam waist** and **Rayleigh range** with **sub-1% agreement** to theory

Experimental Physics II (PC3193):

- Performed **X-ray fluorescence** spectroscopy on brass via **Gaussian deconvolution**; resolved the Cu K_β / Zn K_α overlap; estimated Cu/Zn/Pb mass fractions over five trials
- Indexed **Laue diffraction** patterns for a LiF single crystal; identified Si (diamond-cubic, $a = 5.430 \pm 0.002$ Å) and Mo (BCC, $a = 3.140 \pm 0.004$ Å) via **powder X-ray crystallography**
- Used **Raman spectroscopy** to characterize carbon allotropes; estimated graphene layer count from 2D/G intensity ratios; identified an unknown wafer as crystalline Si from its 521 cm^{-1} mode
- Measured **Paschen curves** for air; extracted the **second Townsend coefficient** ($\gamma = 0.0017 \pm 0.0003$); identified the avalanche-to-streamer breakdown transition
- Ran **TRIM Monte Carlo simulations** of ion transport (nuclear microscopy, implantation, sputtering, proton therapy); analyzed range, straggling, damage, and sputter yields

Biophysics II (PC3267):

- Tracked **protein unfolding** in BSA and lysozyme via **fluorescence spectroscopy**; observed a 24% intensity drop and 19 nm blue shift in BSA; correlated with disulfide-bond count
- Used **circular dichroism** to follow secondary-structure loss under acid denaturation; BSA lost 21% ellipticity versus under 1% for lysozyme, consistent with its four stabilizing disulfides
- Imaged plant and animal cells via **fluorescence microscopy** (Hoechst nuclear staining); compared brightfield and epi-illumination contrast

COMPUTATIONAL PROJECTS

Self-directed unless noted. Interactive notebooks at parthbhargava.net.

Causal Inference in Dynamical Systems *Julia, Python*

2026

Quantified directional information flow with three **transfer entropy** estimators (histogram, KSG, symbolic); recovered coupling direction in coupled Hénon and Lorenz systems via **convergent cross-mapping** (Takens delay-embedding); computed **causal emergence** (ΔEI) across all 256 elementary cellular automata under majority-vote coarse-graining.

Physics Discovery from Data *Julia, Python*

2026

Implemented three law-extraction methods on shared benchmarks: **SINDy** (sparse regression) for ODE identification, **kernel conservation-law discovery** (RBF eigenproblem) for invariants without an assumed functional form, and **genetic-programming symbolic regression**; tested on Van der Pol, pendulum, and Kepler systems.

Spin-Glass Physics of Neural Networks *Julia, Python*

2026

Tracked the **diagonal Hessian curvature spectrum** and **inverse participation ratio** of a small MLP through training, observing a glass-to-order transition as sharp curvature directions emerge, connecting training dynamics to replica-theory ideas for disordered systems.

Coupled Oscillators: Synchronization & Chimera States *Julia, Python*

2026

Simulated Kuramoto dynamics on all-to-all, Erdős-Rényi, and scale-free networks, recovering the threshold $K_c = 2\sigma\sqrt{2/\pi}$ analytically and numerically; extended to Kuramoto-Sakaguchi nonlocal coupling to reproduce **chimera states** and mapped the (K, R) phase plane.

Active Matter: Vicsek Flocking & Run-and-Tumble *Julia, Python*

2026

Simulated the **Vicsek model**, sweeping noise to locate the disorder-to-order nonequilibrium phase transition in the order parameter φ ; modeled **run-and-tumble** swimmers and validated the ballistic-to-diffusive MSD crossover against the analytic effective diffusion constant.

Driven Quadruple Pendulum *Computational Methods (PC3236), with S. Bhar · MATLAB/Python*

2026

Derived the Lagrangian equations of motion for four coupled pendulums under periodic driving; built a custom **RK4 integrator** (validated against scipy RK45 to $< 10^{-6}$ rad over 20 s) and produced phase portraits, Poincaré sections, and bifurcation diagrams tracing the onset of chaos.

Additional self-directed studies: quantum wavepacket evolution (Crank-Nicolson), reservoir computing on chaotic attractors, Gray-Scott reaction-diffusion / Turing patterns, percolation on simplicial complexes, Gō-model protein folding, differentiable parameter inference, and physics-constrained CD spectral inversion.

SKILLS & COURSEWORK

- **Computation:** Python, Julia, MATLAB; numerical methods (Runge-Kutta, Newton-Raphson, Monte Carlo, finite differences); ODE/PDE solvers; data analysis and visualization
- **Core theory:** classical mechanics (Lagrangian/Hamiltonian), electromagnetism, quantum mechanics, thermodynamics and statistical mechanics, mathematical methods (linear algebra, Fourier analysis, complex analysis)
- **Other:** machine learning fundamentals (supervised/unsupervised, neural networks); biophysics (reaction kinetics, self-assembly, molecular dynamics); statistical analysis and uncertainty propagation

RESEARCH INTERESTS

Complex and nonlinear systems, where simple local rules build global structure: collective dynamics, synchronization, networks, and the statistical mechanics of emergence. I am drawn to the line between genuine collective behavior and accumulated noise, and to physics-informed machine learning as a way to probe it.

ACHIEVEMENTS

- **Silver Medal**, International Aerospace Olympiad 2024
- **JEE Mains:** 99.14 percentile (top 1% of 1.5 million candidates)